

ANNEXURE A — ASSESSMENT SOLUTIONS

Question 1

Question

A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them. One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP, FP, FN, TN.

Given Data

- **Actual Condition (Wolf State):** Positive = Real wolf present; Negative = No wolf present.
- **Prediction / System Alarm:** Positive = Alarm raised ("Wolf!"); Negative = No alarm / Alarm ignored (Villagers stay home).

Problem Solving Analysis

- **False Positive (FP):** Predicting positive when actual is negative $\rightarrow$ The boy shouts "Wolf!" when there is *no wolf* (False alarm).
- **False Negative (FN):** Predicting negative when actual is positive $\rightarrow$ The villagers stay in their houses (assuming "No Wolf") when a *real wolf arrives* (Missed detection).
- **True Negative (TN):** Predicting negative when actual is negative $\rightarrow$ Peaceful nights with *no wolf present* and *no alarm acted upon*.
- **True Positive (TP):** Predicting positive when actual is positive $\rightarrow$ The boy shouting "Wolf!" when a *real wolf arrives* (Correct detection signal by the boy).

Output

- **True Positive (TP):** Boy shouting "Wolf!" when the real wolf arrived.
- **False Positive (FP):** Boy crying "Wolf!" when no wolf was present.
- **False Negative (FN):** Villagers staying in their houses when a real wolf arrived.
- **True Negative (TN):** Peaceful state with no wolf and no alarm raised.

Question 2

Question

Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP, FP, FN, TN and Calculate the Precision, Recall, and F1 Score.

Given Data

- Total images (\$N\$) = 100
- Actual Cat images (\$P\$) = 30
- Actual Non-Cat images (\$N\$) = 100 - 30 = 70
- Predicted Cat in Cat images = 25
- Predicted Non-Cat in Non-Cat images = 50

Problem Solving Analysis

Confusion Matrix Values:

- **TP:** Cat images correctly predicted as Cat = **25**
- **FN:** Cat images incorrectly predicted as Non-Cat = \$30 - 25 = \$ **5**
- **TN:** Non-Cat images correctly predicted as Non-Cat = **50**
- **FP:** Non-Cat images incorrectly predicted as Cat = \$70 - 50 = \$ **20**

Metric Calculations:

- $Precision = \frac{TP}{TP + FP} = \frac{25}{25 + 20} = \frac{25}{45} = \frac{5}{9} \approx 0.5556 \text{ \textit{quad} (55.56\%)}$
- $Recall = \frac{TP}{TP + FN} = \frac{25}{25 + 5} = \frac{25}{30} = \frac{5}{6} \approx 0.8333 \text{ \textit{quad} (83.33\%)}$
- $F1 \text{ ext{ Score} } = 2 \text{ imes } \frac{\text{ext{Precision}} \text{ imes ext{Recall}}}{\text{ext{Precision}} + \text{ext{Recall}}} = 2 \text{ imes } \frac{\frac{5}{9} \text{ imes } \frac{5}{6}}{\frac{5}{9} + \frac{5}{6}} = 2 \text{ imes } \frac{1}{3} = \frac{2}{3} \approx 0.6667 \text{ \textit{quad} (66.67\%)}$

Output

- **TP:** 25, **FP:** 20, **FN:** 5, **TN:** 50
- **Precision:** 0.5556 (55.56%)
- **Recall:** 0.8333 (83.33%)
- **F1 Score:** 0.6667 (66.67%)

Question 3

Question

An email service provider wants to classify emails as Spam or Not Spam using features: Contains "Offer" (Yes/No), Contains "Free" (Yes/No), Email Length (Short/Long).

Sample Dataset:

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions:

1. Calculate prior probabilities: $P(\text{ext}\{\text{Spam}\})$, $P(\text{ext}\{\text{Not Spam}\})$.
2. Compute conditional probabilities for each feature.
3. Using Naïve Bayes, classify: $P(\text{ext}\{\text{Offer}\}=\text{ext}\{\text{Yes}\}, \text{ext}\{\text{Free}\}=\text{ext}\{\text{No}\}, \text{ext}\{\text{Length}\}=\text{ext}\{\text{Short}\})$.
4. Compare results using WEKA (Naïve Bayes classifier).
5. Discuss the independence assumption in Naïve Bayes.

Given Data

- Total Instances (N) = 6 | Spam = 3, Not Spam = 3
- Test Instance X : $P(\text{ext}\{\text{Offer}\}=\text{ext}\{\text{Yes}\}, \text{ext}\{\text{Free}\}=\text{ext}\{\text{No}\}, \text{ext}\{\text{Length}\}=\text{ext}\{\text{Short}\})$

Problem Solving Analysis

1. Prior Probabilities:

- $P(\text{ext}\{\text{Spam}\}) = 3/6 = 0.50$
- $P(\text{ext}\{\text{Not Spam}\}) = 3/6 = 0.50$

2. Conditional Probabilities:

- Class = Spam (3 samples):
 $P(\text{ext}\{\text{Offer}\}=\text{ext}\{\text{Yes}\}|\text{ext}\{\text{Spam}\}) = 2/3$, $P(\text{ext}\{\text{Offer}\}=\text{ext}\{\text{No}\}|\text{ext}\{\text{Spam}\}) = 1/3$
 $P(\text{ext}\{\text{Free}\}=\text{ext}\{\text{Yes}\}|\text{ext}\{\text{Spam}\}) = 3/3 = 1.0$, $P(\text{ext}\{\text{Free}\}=\text{ext}\{\text{No}\}|\text{ext}\{\text{Spam}\}) = 0/3 = 0.0$
 $P(\text{ext}\{\text{Length}\}=\text{ext}\{\text{Short}\}|\text{ext}\{\text{Spam}\}) = 2/3$, $P(\text{ext}\{\text{Length}\}=\text{ext}\{\text{Long}\}|\text{ext}\{\text{Spam}\}) = 1/3$
- Class = Not Spam (3 samples):
 $P(\text{ext}\{\text{Offer}\}=\text{ext}\{\text{Yes}\}|\text{ext}\{\text{Not Spam}\}) = 1/3$, $P(\text{ext}\{\text{Offer}\}=\text{ext}\{\text{No}\}|\text{ext}\{\text{Not Spam}\}) = 2/3$
 $P(\text{ext}\{\text{Free}\}=\text{ext}\{\text{Yes}\}|\text{ext}\{\text{Not Spam}\}) = 0/3 = 0.0$, $P(\text{ext}\{\text{Free}\}=\text{ext}\{\text{No}\}|\text{ext}\{\text{Not Spam}\}) = 3/3 = 1.0$
 $P(\text{ext}\{\text{Length}\}=\text{ext}\{\text{Short}\}|\text{ext}\{\text{Not Spam}\}) = 1/3$, $P(\text{ext}\{\text{Length}\}=\text{ext}\{\text{Long}\}|\text{ext}\{\text{Not Spam}\}) = 2/3$

3. Naïve Bayes Classification for $X = (\text{Offer}=\text{Yes}, \text{Free}=\text{No}, \text{Length}=\text{Short})$:

- $P(\text{ext}\{\text{Spam}\}|X) \propto P(\text{ext}\{\text{Spam}\}) \times P(\text{ext}\{\text{Offer}\}=\text{ext}\{\text{Yes}\}|\text{ext}\{\text{Spam}\}) \times P(\text{ext}\{\text{Free}\}=\text{ext}\{\text{No}\}|\text{ext}\{\text{Spam}\}) \times P(\text{ext}\{\text{Length}\}=\text{ext}\{\text{Short}\}|\text{ext}\{\text{Spam}\})$
 $= 0.50 \times \frac{2}{3} \times 0.0 \times \frac{2}{3} = \mathbf{0.0}$
- $P(\text{ext}\{\text{Not Spam}\}|X) \propto P(\text{ext}\{\text{Not Spam}\}) \times P(\text{ext}\{\text{Offer}\}=\text{ext}\{\text{Yes}\}|\text{ext}\{\text{Not Spam}\}) \times P(\text{ext}\{\text{Free}\}=\text{ext}\{\text{No}\}|\text{ext}\{\text{Not Spam}\}) \times P(\text{ext}\{\text{Length}\}=\text{ext}\{\text{Short}\}|\text{ext}\{\text{Not Spam}\})$
 $= 0.50 \times \frac{1}{3} \times 1.0 \times \frac{1}{3} = \frac{1}{18} \approx \mathbf{0.0556}$
- Normalized: $P(\text{ext}\{\text{Spam}\}|X) = 0\%$, $P(\text{ext}\{\text{Not Spam}\}|X) = 100\%$ $\rightarrow$ Predicted Class: **Not Spam**.

4. WEKA Comparison:

Dataset in ARFF format for WEKA:

```

@relation email_spam
@attribute Offer {Yes, No}
@attribute Free {Yes, No}
@attribute Length {Short, Long}
@attribute Class {Spam, Not_Spam}
@data
Yes, Yes, Short, Spam
Yes, No, Long, Not_Spam
No, Yes, Short, Spam
No, No, Long, Not_Spam
Yes, Yes, Long, Spam
No, No, Short, Not_Spam
Yes, No, Short, ?

```

WEKA's NaiveBayes classifier predicts **Not_Spam** for the test instance, which perfectly matches our mathematical solution.

5. Independence Assumption in Naïve Bayes:

Naïve Bayes assumes that attribute values are conditionally independent given the class label: $P(X_1, X_2, \dots, X_n | Y) = \prod P(X_i | Y)$. In real-world datasets, features are often correlated (e.g., "Offer" and "Free"). Although this violation distorts probability magnitude estimates, Naïve Bayes still yields accurate classification rankings.

Output

- **1. Prior Probabilities:** $P(\text{Spam}) = 0.50$, $P(\text{Not Spam}) = 0.50$
- **2. Conditional Probabilities:** Calculated for all features per class.
- **3. Classification Result:** **Not Spam** ($P(\text{Not Spam} | X) = 100\%$)
- **4. WEKA Comparison:** Matches WEKA prediction (**Not_Spam**).
- **5. Independence Assumption:** Explained conditional independence and why ranking remains robust despite feature correlation.